

»» Chapter VIII ««

THE ZEROS AND POLES OF MEROMORPHIC FUNCTIONS

PREFATORY REMARKS

The basic problem of this chapter is that of the distribution of the zeros and poles of a meromorphic function. Specifically, given a region $A \subset \hat{\mathbb{C}}$ we wish to characterize the divisors of the functions meromorphic on A and to determine the extent to which the principal parts at the poles may be specified. When $A = \hat{\mathbb{C}}$, we are practically in full possession of the facts. As we have seen in §4, Ch. VI, if $f \in \mathbf{M}(\hat{\mathbb{C}})$ and f is not the constant 0, there are a finite number of zeros and poles and the sum of the divisors of f over the set of these points is 0. The stated property is characteristic. If $\partial: \hat{\mathbb{C}} \rightarrow \mathbb{Z}$ is such that $\{\partial(z) \neq 0\}$ is finite and the sum of ∂ over this set is 0, then there exists $f \in \mathbf{M}(\hat{\mathbb{C}})$ such that $\partial_f = \partial$, and indeed f is determined up to a nonzero constant factor. To see this, suppose that n is the sum of ∂ over $\{\partial(z) > 0\}$. We proceed by induction. When $n = 0$, the assertion is trivial, the f being the nonzero constants of $\mathbf{M}(\hat{\mathbb{C}})$. If the sum in question is $n + 1$, $n \in \mathbb{N}$, let $\partial(a) > 0$, $\partial(b) < 0$, and let m denote a Möbius transformation satisfying: $m(a) = 0$, $m(b) = \infty$. Then $\partial - \partial_m$ satisfies the stated conditions with sum in question now being n rather than $n + 1$. Hence with f satisfying $\partial_f = \partial - \partial_m$, we have $\partial_{fm} = \partial$. The remainder of the assertion is immediate. As far as principal parts are concerned, it is clear that if $a_1, \dots, a_n$ are n distinct points of $\hat{\mathbb{C}}$, and $P_k \in \mathbf{M}(\hat{\mathbb{C}})$ has a sole pole, in fact at a_k , then there exist $f \in \mathbf{M}(\hat{\mathbb{C}})$, analytic in $\hat{\mathbb{C}} - \{a_1, \dots, a_n\}$ and having the same principal part at a_k as

P_k , for example, $\sum_1^n P_k$, and the difference of such functions is a constant. Thus the totality consists of the functions of the form $c + \sum_1^n P_k$, c constant.

When we turn to a region A , a *proper* part of $\hat{\mathbb{C}}$, the situation changes radically. Here we have to cope with the possibility of infinitely many zeros and poles. If $f \in M(A)$ is not the constant zero, then the zeros of f do not cluster at a point of A . We conclude that ∂_f takes values in $\mathbb{Z}$ and $\{\partial_f(z) \neq 0\}$ clusters at no point of A . The general existence theorem of Weierstrass (§1, this chapter) implies that a sufficient condition that $\partial: A \rightarrow \mathbb{Z}$ be the divisor of a member of $M(A)$, not the constant zero, is that $\{\partial(z) \neq 0\}$ cluster at no point of A . The Mittag-Leffler theorem (§1, this chapter) implies that if $B \subset A$ clusters at no point of A , then there exists $f \in M(A)$ having an assigned principal part at each point of B and analytic in $A - B$. If f is such a function, then the totality of such functions consists of the functions of the form $f + g$, g analytic on A . In these theorems A is replaced by an arbitrary proper nonempty open subset of $\hat{\mathbb{C}}$.

The two cited theorems are the coarse qualitative results of the theory. They lead to important conclusions. We shall meet the following applications of the Weierstrass theorem: (1) When $A \neq \hat{\mathbb{C}}$, $M(A)$ is the field generated by $A(A)$ = the ring of functions analytic on A . (2) If θ is an isomorphism of $A(A)$ onto $A(B)$, A and B being regions which are proper parts of $\hat{\mathbb{C}}$, and θ "preserves" the constant taking the value i , then there exists a unique univalent map φ of B onto A , meromorphic on B such that $\theta(f) = f \circ \varphi$, $f \in A(A)$. A corresponding assertion holds when θ satisfies the alternative condition of taking the constant with value i into the constant with value $-i$. This striking theorem (2) was given by L. Bers [135]. We shall give a proof due to Nakai [144]. The cited paper of Nakai extends the theorem of Bers to the setting of Riemann surface theory. Recently, great advances were made by H. Iss'sa [138] who obtained the corresponding theorem for field isomorphisms. The resources of his study go beyond the program of this book.

We have emphasized the *qualitative* character of the theorems of Weierstrass and Mittag-Leffler. When conditions of a quantitative nature (growth conditions) are imposed and we restrict our attention to special domains, more refined results emerge. It is intentional that we introduce infinite products of complex numbers and complex-valued functions just before turning to the more refined problems. No special advantage is gained by treating infinite products earlier, and one might even say that the special restrictions in the theory of infinite products become significant when viewed in terms of limit questions of the kind encountered in the proof of the Weierstrass theorem.

We shall consider in the latter part of the present chapter the distribution of the zeros of a function analytic on a disk or a half-plane and bounded, as well as the distribution of the zeros of an entire function f satisfying

$|f(z)| \leq A \exp(|z|^B)$, $z \in \mathbb{C}$, A and B being positive numbers. These are questions where growth conditions are present.

1. THE THEOREMS OF MITTAG-LEFFLER AND WEIERSTRASS

We shall treat the Mittag-Leffler theorem first. It states

Theorem 1.1: *Let O be a nonempty proper open subset of $\hat{\mathbb{C}}$, let $B \subset O$ cluster at no point of O , let there be assigned to each $b \in B$ a function $g_b \in \mathbf{M}(\hat{\mathbb{C}})$ having a pole at b and no others. Then there exists $f \in \mathbf{M}(O)$, analytic in $O - B$ and having the same principal part at b as g_b . If f is such a function, the totality of such functions consists of the functions $f + g$, g analytic on O .*

PROOF: We note first that we may assume that $\infty \in O - B$. To see that this is so, we let $a \in O - B$ and let m denote a Möbius transformation satisfying $m(a) = \infty$. If the restricted theorem is valid, then there exists $f \in \mathbf{M}[m(O)]$, analytic in $m(O - B)$ and satisfying the condition that for each $b \in B$ the principal part of f at $m(b)$ is equal to that of $g_b \circ m^{-1}$. Hence $f \circ (m|O) \in \mathbf{M}(O)$, its poles are the points of B and since for given $b \in B$, $f - (g_b \circ m^{-1})|m(O)$ is analytic at $m(b)$, we conclude that $f \circ (m|O) - g_b|O$ is analytic at b and $f \circ (m|O)$ has the desired property of having the same principal part at b as g_b . We proceed, as we may, assuming that $\infty \in O - B$. We also assume that B is infinite, for the case where B is finite is immediate, $\sum_{b \in B} (g_b|O)$ yielding a function of the desired type.

Given $b \in B$, there exists a point $c \in \mathbb{C} - O$ such that

$$|b - c| = \text{dist}(\{b\}, \mathbb{C} - O)$$

since $\mathbb{C} - O$ is nonempty and compact. Now g_b is analytic in $\hat{\mathbb{C}} - \overline{\Delta(c; r)}$, $r = |b - c|$. Considering the Laurent expansion of g_b in $\{r < |z - c| < +\infty\}$, we see that, given ε positive, there exists $h \in \mathbf{M}(\hat{\mathbb{C}})$ having a pole at most at c and such that

$$|g_b(z) - h(z)| < \varepsilon, \quad |z - c| > 2r.$$

Now let $(b_n)_{n \in \mathbb{N}}$ denote a univalent enumeration of B . It is to be noted that given δ positive the set of $b \in B$ satisfying $\text{dist}(\{b\}, \mathbb{C} - O) \geq \delta$ is compact, and, since it clusters at no point, it is finite. Let c_n and r_n have the same context relative to b_n as do c and r relative to b in the preceding paragraph, and let h_n take over the role of h in the present context where $b = b_n$, $c = c_n$, $r = r_n$, and ε is taken as 2^{-n} . We propose to establish the desired existence assertion

by verifying that

$$\sum_0^{\infty} (g_{b_n} - h_n) \mid O \quad (1.1)$$

has the required properties. From the observation of the second sentence of this paragraph we see that $\lim_{n \rightarrow \infty} r_n = 0$. Now suppose that $\overline{\Delta(a; \rho)} \subset O$, $a \in \mathbb{C}$, and note that for n large we have

$$2r_n < \text{dist}(\overline{\Delta(a; \rho)}, \mathbb{C} - O).$$

Hence (1.1) converges χ -uniformly on $\Delta(a; \rho)$ since

$$|g_{b_n}(z) - h_n(z)| < 2^{-n}, \quad (1.2)$$

$z \in \Delta(a; \rho)$, for n sufficiently large. It is to be noted that (1.1) converges uniformly (in the sense of **C**) on $\Delta(\infty; \rho)$ when $1/\rho > |c_n| + 2r_n$, $n \in \mathbb{N}$. We conclude that (1.1) converges χ -uniformly in O and has a meromorphic sum f (cf. Ex. 4.35, Ch. VI). It is clear that f has the same principal part as g_b at b . The last sentence of the theorem is immediate.

EXERCISE

1.1. Suppose that g_b satisfies instead of the stated condition the requirement that g_b be analytic on $\hat{\mathbb{C}} - \{b\}$. Prove the existence of f analytic on $O - B$ such that f and g_b have the same principal part at each point of B .

The Weierstrass theorem will be formulated as follows.

Theorem 1.2: Given O a proper nonempty open subset of $\hat{\mathbb{C}}$. Let $\partial: O \rightarrow \mathbb{Z}$ be such that $\{\partial(z) \neq 0\}$ clusters at no point of O . Then there exists $f \in \mathbf{M}(O)$ such that $\partial_f = \partial$. Further the totality of such functions consists of the functions fg where g is analytic on O and has no zeros.

PROOF: We suppose that $\infty \in O$ and $\partial(\infty) = 0$, for we may employ the reduction used in the Mittag-Leffler theorem, considering $a \in O$ such that $\partial(a) = 0$, and $F \in \mathbf{M}[m(O)]$ satisfying $\partial_F \circ (m \mid O) = \partial$. We suppose that $B = \{\partial(z) \neq 0\}$ is infinite, for when B is finite, it suffices to let $w \in \hat{\mathbb{C}} - O$ and to introduce a member of $\mathbf{M}(\hat{\mathbb{C}})$ whose divisor takes the value 0 in the complement of $B \cup \{w\}$ with respect to $\hat{\mathbb{C}}$ and takes the value $\partial(z)$ at $z \in B$ and $-d$ at w , where d is the sum of ∂ over B . We continue to use the notation of the proof of the Mittag-Leffler theorem.

It will be convenient to introduce E_p , the *Weierstrass primary factor* with

index $p \in \mathbb{N}$, given by

$$E_p(z) = (1 - z) \exp\left(\sum_1^p \frac{z^k}{k}\right). \quad (1.3)$$

Clearly, E_p has a simple zero at 1 and no others. A special feature of E_p is the fact that $\lambda_p = -\sum_{p+1}^{\infty} z^k/k$ with domain $\Delta(0; 1)$ is an analytic logarithm of $E_p | \Delta(0; 1)$.

For each $n \in \mathbb{N}$ let $m(n) \in \mathbb{N}$ be so chosen that

$$|\partial(b_n)| \sum_{m(n)+1}^{\infty} \frac{|z|^k}{k} < 2^{-n}, \quad |z| \leq 2^{-1}.$$

We consider the sequence (F_n) of functions meromorphic on O where

$$F_n = \prod_0^n (E_{m(k)} \circ \mu_k)^{\partial(b_k)}.$$

Here

$$\mu_k(z) = \frac{b_k - c_k}{z - c_k}, \quad z \in O.$$

Taking $\Delta(a; \rho)$ as in the proof of the Mittag-Leffler theorem and taking $q \in \mathbb{N}$ such that

$$\mu_k[\Delta(a; \rho)] \subset \Delta(0; \tfrac{1}{2}), \quad k \geq q + 1,$$

we see that

$$\sum_{q+1}^{\infty} \partial(b_k) \lambda_{m(k)} \circ [\mu_k | \Delta(a; \rho)]$$

converges uniformly in the sense of the Euclidean metric of $\mathbb{C}$ and has a bounded sum. Since

$$\sum_{q+1}^{q+n+1} \partial(b_k) \lambda_{m(k)} \circ [\mu_k | \Delta(a; \rho)]$$

is an analytic logarithm of $(F_{q+n+1}/F_q) | \Delta(a; \rho)$, we conclude that the sequence whose n th term is $(F_{q+n+1}/F_q) | \Delta(a; \rho)$ tends uniformly to an analytic function with no zero.

The proof of the theorem is now readily completed. We see that (F_n) tends χ -uniformly in O to a meromorphic function with divisor ∂ . It is to be noted that since $\bigcup_{n \in \mathbb{N}} \Delta(c_n; 2r_n)$ is bounded, $(F_n | \Delta(\infty; \rho))$ tends uniformly to a limit not taking the value zero for some positive ρ .

The last assertion of the theorem is immediate.

EXERCISES

1.2. It is easy to construct a sequence (f_n) of functions taking values in $\mathbb{C}$ which tends uniformly to its limit in the sense of the metric of $\mathbb{C}$ but is such that $(\exp \circ f_n)$ does not tend uniformly to $\exp \circ f$. Example: $f_n = x^{-1} + (n+1)^{-1}$, $0 < x \leq 1$, $n \in \mathbb{N}$. Validate the assertion of the text concerning the uniform limit property of $((F_{q+n+1}/F_q) | \Delta(a; \rho))$.

1.3. Show that when $p > 0$, there exists a positive number A such that

$$\begin{aligned} |E_p(z)| &\leq \exp(A|z|^{p+1}), & |z| \leq 1; \\ &\leq \exp(A|z|^p), & |z| > 1. \end{aligned}$$

Hint: Consider $[E_p(z) - 1]z^{-(p+1)}$ on $\overline{\Delta(0; 1)}$, the appropriate definition being made at 0. Make a direct estimate of $|E_p(z)|$ when $|z| > 1$.

1.4. Given $g \in \mathbf{M}(\hat{\mathbb{C}})$ having a sole pole at $b \in \mathbb{C} - \{0\}$, $g(\infty) = 0$, and $n \in \mathbb{N}$, we define

$$T_n(g) = g(z) - \sum_0^n \frac{g^{(k)}(0) z^k}{k!}, \quad z \in \mathbb{C}.$$

Suppose that B is an infinite subset of $\mathbb{C}$ clustering at no point of $\mathbb{C}$ and that for each $b \in B$ there is given g_b having a sole pole at b . Let $(b_n)_{n \in \mathbb{N}}$ denote a univalent enumeration of B . Show that there exists $\nu: \mathbb{N} \rightarrow \mathbb{N}$ such that $\sum_0^\infty T_{\nu(k)}(g_{b_k})$ converges χ -uniformly in $\mathbb{C}$ to a meromorphic sum with poles in B , the principal part at b being given by g_b .

Carry out the corresponding program for the Weierstrass theorem using products

$$\prod_0^n \left[E_{\nu(k)} \left(\frac{z}{b_k} \right) \right]^{o(b_k)}$$

1.5. Let O be an open nonempty proper subset of $\hat{\mathbb{C}}$. Show that there exists $(c_n): \mathbb{N} \rightarrow O$ univalent such that $\text{fr } O$ is the set of cluster points of $\{c_n: n \in \mathbb{N}\}$. Show that there exists f analytic on O such that $\partial_f(c_n) = 1$, $n \in \mathbb{N}$, and conclude that f is not the restriction to O of a function meromorphic (let alone analytic) at each point of O and at some point of $\text{fr } O$.

1.6. Let A be a region that is a proper subset of $\hat{\mathbb{C}}$. Show that $\mathbf{M}(A)$ is the field generated by $\mathbf{A}(A)$. Hint: Given $f \in \mathbf{M}(A)$ introduce $g \in \mathbf{A}(A)$ satisfying $\partial_g = -\partial_f^-$.

***1.7.** Let A be a region in $\hat{\mathbb{C}}$. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of functions meromorphic on A , none the constant zero, satisfying: (1) there exists $a \in A$ such that $(f_n(a))$ possesses a limit in $\mathbb{C} - \{0\}$, (2) (f_n'/f_n) tends χ -uniformly in A to a function g meromorphic on A and satisfying $\partial_g(z) \geq -1$, $z \in A$, $\partial_g(a) \geq 0$. Show that (f_n) tends χ -uniformly in A to a function F meromorphic on A which satisfies $f'/f = g$ (*) and that the totality of functions meromorphic on A and satisfying (*) consists of the nonzero constant multiples of F . Hint: Apply a connectedness argument introducing E , the set of $z \in A$ for which there exists O open, $z \in O \subset A$, such that $(f_n | O)$ tends χ -uniformly to a function meromorphic on O , not the constant zero. What happens when the hypothesis on ∂_g is dropped?

Show by example that when (f_n) satisfies instead of (2) the condition that (f_n) tend χ -uniformly in A to f meromorphic on A , it is not necessarily the case that (f_n'/f_n)

tends χ -uniformly in A to f'/f . What is a necessary and sufficient condition on such (f_n) and f for (f'_n/f_n) to tend χ -uniformly in A to f'/f ?

1.8. Apply Ex. 1.7 to obtain from the expansion of $\pi \cot \pi z$ given in §3, Ch. VII,

$$\sin \pi z = \lim_{n \rightarrow \infty} \pi z \prod_{k=1}^n (1 - k^{-2} z^2).$$

2. THE ISOMORPHISM THEOREM OF BERS

Thanks to the Weierstrass theorem, Theorem 1.2, this chapter, we are in a position to give an account of Nakai's proof of the theorem of Bers. In what follows A and B are regions in $\hat{\mathbb{C}}$, θ is an isomorphism of $\mathbf{A}(A)$ onto $\mathbf{A}(B)$; that is, θ is a univalent map of (AA) onto $\mathbf{A}(B)$ satisfying

$$\theta(f + g) = \theta(f) + \theta(g), \quad \theta(fg) = \theta(f)\theta(g), \quad f, g \in \mathbf{A}(A). \quad (2.1)$$

The theorem of Bers states

Theorem 2.1: *If $A \neq \hat{\mathbb{C}}$, then either there exists a unique $\varphi: B \rightarrow A$ such that*

$$\theta(f) = f \circ \varphi, \quad f \in \mathbf{A}(A),$$

or else there exists a unique $\varphi: B \rightarrow A$ such that

$$\theta(f) = \bar{f} \circ \varphi, \quad f \in \mathbf{A}(A).$$

Further φ is a univalent map of B onto A , meromorphic on B in the former case, and such that the conjugate of φ is meromorphic on B in the latter case.

PROOF: It is understood that by the *conjugate* $\bar{f}$ of a function f , taking values in $\hat{\mathbb{C}}$ is meant $\alpha \circ f$, where $\alpha(z) = \bar{z}$, $z \in \mathbb{C}$, $\alpha(\infty) = \infty$. The *conjugation* map α is clearly a homeomorphism of $\hat{\mathbb{C}}$ onto itself.

We start the proof by studying the behavior of constant functions under θ . To prevent confusion of meaning we shall introduce special notation for constant functions. Thus given $c \in \mathbb{C}$, by c_A will be meant the function with domain A taking at each point the value c . We see at once from (2.1) that $\theta(0_A) = 0_B$, $\theta(1_A) = 1_B$. It follows by induction from (2.1) that $\theta(n_A) = n_B$, $n \in \mathbb{N}$, and thereupon by standard arguments that $\theta(q_A) = q_B$, $q \in \mathbb{Q}$. Since $[\theta(i_A)]^2 = (-1)_B$, $\theta(i_A)$, which is analytic, is either i_B or $(-i)_B$. Thus either θ maps each constant of $\mathbf{A}$ taking a *rational complex* value (that is, of the form $a + bi$, $a, b \in \mathbb{Q}$) onto the constant of $\mathbf{A}(B)$ taking the same value or θ maps each such constant c_A onto $(\bar{c})_B$. Now a constant is the n th power of a constant, n a positive integer. Hence given $c \in \mathbb{C}$ and n a positive integer, $\theta(c_A)$ is the n th power of a member of $\mathbf{A}(B)$. Hence if $c \neq 0$, $\theta(c_A)$ does not have any zeros. Hence if c is complex but not rational complex and q is

rational complex, $\theta[(c - q)_A]$ does not have a zero. Consequently $\theta(c_A)$ omits all rational complex values. By the open mapping property of nonconstant analytic functions we see that $\theta(c_A)$ is a constant. That θ maps the set of constants of $A(A)$ onto the set of constants of $A(B)$ follows on noting that θ^{-1} maps a constant of $A(B)$ onto a constant of $A(A)$.†

To establish the theorem of Bers we employ the notion of a *maximal principal ideal*, which is a concept of fundamental utility in questions concerning rings of functions. In the case of a commutative ring with identity, $\mathfrak{R}$, a *principal ideal* turns out to be a set of the form $\{ax: x \in \mathfrak{R}\}$, $a \in \mathfrak{R}$.‡ By a *maximal principal ideal* of $\mathfrak{R}$ is meant a principal ideal I properly contained in $\mathfrak{R}$ but not properly contained in a principal ideal J itself properly contained in $\mathfrak{R}$. We proceed by determining the maximal principal ideals of $A(A)$ and the action of θ on them.

Given $a \in A$, $I(a) = \{f \in A(A), f(a) = 0\}$ is a principal ideal of $A(A)$. For let $g \in A(A)$ have a simple zero at a and no other zeros. From $f = (f/g)g$ we see that $I(a)$ is a principal ideal. Suppose that J is a principal ideal, say $\{fh: f \in A(A)\}$, $h \in A(A)$, containing $I(a)$ properly. Now $h(a) \neq 0$. If $f \in A(A)$,

$$f - \frac{f(a)}{h(a)}h \in I(a) \subset J,$$

and hence $f \in J$, so that $A(A) = J$. We conclude that $I(a)$ is a maximal principal ideal.

Let us reverse the question and show that every maximal principal ideal of $A(A)$ is $I(a)$ for a unique $a \in A$. Unicity is easy. $I(a)$ has a member having a as its sole zero. Hence $I(a) = I(b)$ implies that $a = b$. Let $g \in A(A)$ and suppose that $I = \{fg: f \in A(A)\}$. If g has no zero, we have $I = A(A)$, since $h = (h/g)g \in I$, $h \in A(A)$. Hence if $I \neq A(A)$, $g(a) = 0$ for some $a \in A$. Hence $I \subset I(a)$ and we conclude that the maximal principal ideals are precisely the $I(a)$.

The action of θ on $I(a)$. Since θ is an isomorphism mapping $A(A)$ onto $A(B)$, we see that $\theta[I(a)]$ is a principal ideal of $A(B)$ and is a proper part of $A(B)$ as well. Hence by the argument of the preceding paragraph $\theta[I(a)] \subset J(b)$ for some $b \in B$, $J(b)$ denoting $\{g \in A(B), g(b) = 0\}$. From $I(a) \subset \theta^{-1}[J(b)]$, the maximality of $I(a)$, and the fact that $\theta^{-1}[J(b)]$ is a principal ideal of $A(A)$ properly contained in $A(A)$, we conclude that $I(a) = \theta^{-1}[J(b)]$ or, equivalently, $\theta[I(a)] = J(b)$. We introduce ψ , the set of ordered pairs $(a, b) \in A \times B$ satisfying

† If we abandon for the moment the hypothesis $A \neq \hat{C}$, and suppose on the contrary that $A = \hat{C}$, we see that $A(B)$ consists of constants. Since $A(B)$ has nonconstant members if $B \neq \hat{C}$, for example, $m|B$ where m is a Möbius transformation taking some point of $\hat{C} - B$ into ∞ , we see that $B = \hat{C}$. In this case, θ may be extremely complicated.

‡ The unlikely reader unfamiliar with the notion of an ideal is to take the preceding sentence as the basis for a stopgap definition.

$\theta[I(a)] = J(b)$ and show that ψ is a function and maps A univalently onto B . All further developments depend upon the properties of ψ . Clearly,

$$\{a: (a, b) \in \psi\} = A,$$

and, since $J(b) = J(c)$ implies $b = c$, ψ is a function with domain A . If $\psi(a_1) = \psi(a_2)$, then $I(a_1) = I(a_2)$ and hence $a_1 = a_2$. Applying the above reasoning to θ^{-1} we conclude that given $b \in B$ we have $\theta^{-1}[J(b)] = I(a)$ for some $a \in A$, so that $b = \psi(a)$. Hence $\psi(A) = B$.

We note that $f - f(a)_A \in I(a)$. From $\theta[f - f(a)_A] \in J[\psi(a)]$, we conclude

$$\theta(f)(\psi(a)) = \theta[f(a)_A](\psi(a)). \quad (2.2)$$

If $\theta(f)(\psi(a))$ is rational complex, we have $\theta(f)(\psi(a)) = f(a)$ in case $\theta(i_A) = i_B$ and $\theta(f)(\psi(a)) = \overline{f(a)}$ in case $\theta(i_A) = -i_B$.

To continue our study of ψ we consider $(a_n): \mathbb{N} \rightarrow A$ univalent and such that $\{\psi(a_n): n \in \mathbb{N}\}$ clusters at no point of B . Let $g \in A(B)$ and satisfy $\partial_g[\psi(a_n)] = 1$, $n \in \mathbb{N}$, the existence of such g being assured by the Weierstrass theorem of the preceding section. Let $f = \theta^{-1}(g)$. Clearly $f \neq 0_A$ and $f(a_n) = 0$, $n \in \mathbb{N}$. Hence $\{a_n: n \in \mathbb{N}\}$ clusters at no point of A . Obviously, ψ^{-1} enjoys the same property, *mutatis mutandis*.

We next show: if $a \in A$, then $\lim_a \psi$ exists and is in B . Indeed, there exist r positive and K compact $\subset B$ such that $\Delta(a; r) \subset A$, $\psi[\Delta(a; r)] \subset K$. Otherwise there would exist $(a_n): \mathbb{N} \rightarrow A$ univalent and such that $\lim_{n \rightarrow \infty} a_n = a$ but $\{\psi(a_n): n \in \mathbb{N}\}$ clusters at no point of B .† Hence if $\lim_a \psi$ does not exist, there are two univalent sequences (a_n) , (α_n) with elements in A and limit a such that (b_n) and (β_n) , where $b_n = \psi(a_n)$ and $\beta_n = \psi(\alpha_n)$, possess limits in B , say b and β , respectively, $b \neq \beta$. We suppose, as we may, that $a_m \neq \alpha_n$, $m, n \in \mathbb{N}$, and that $a \neq a_n$, $a \neq \alpha_n$, $b_n \neq b$, $b_n \neq \beta$, $\beta_n \neq b$, $\beta_n \neq \beta$ for each $n \in \mathbb{N}$.

We shall see in the next paragraph that the conditions on b , b_n , β , β_n imply that there exists $g \in A(B)$ satisfying: $g(b) \neq g(\beta)$; $g(b_n)$ and $g(\beta_n)$ are rational complex, $n \in \mathbb{N}$. With $f = \theta^{-1}(g)$, by the remark following (2.2) either $f(a_n) = g(b_n)$ and $f(\alpha_n) = g(\beta_n)$, $n \in \mathbb{N}$, or else $f(a_n) = \overline{g(b_n)}$ and $f(\alpha_n) = \overline{g(\beta_n)}$, $n \in \mathbb{N}$. In either case the continuity of f at a is violated. Hence $\lim_a \psi$ exists, and using the compactness of K we see that $\lim_a \psi \in B$.

Existence of g . For each $n \in \mathbb{N}$ we introduce auxiliary functions u_n and w_n in $A(B)$ satisfying: $u_n(b) = 0$, $u_n(b_k) = 0$ for $k = 0, \dots, n-1$, $u_n(b_n) = 1$,

† To obtain (a_n) we introduce

$$K = \bigcap_{\epsilon > 0} \{\chi(z, \epsilon) \geq r\}$$

and take a sequence of positive r with limit 0. The Riemann surface situation is treated differently, but the point is not a serious one.

$u_n(\beta_k) = u_n(\beta) = 0$ for $k = 0, \dots, n$; $w_n(b_k) = w_n(b) = 0$ for $k = 0, \dots, n$, $w_n(\beta) = 0$, $w_n(\beta_k) = 0$ for $k = 0, \dots, n-1$, $w_n(\beta_n) = 1$. Let $h \in A(B)$ and satisfy $h(b) = 0$, $h(\beta) = 1$. We seek to construct g as a function of the form

$$h + \sum_0^\infty (\gamma_n u_n + \delta_n w_n)$$

where $\gamma_n, \delta_n \in \mathbb{C}$ are to be so chosen that uniform convergence in B to an analytic function is assured and that the function g satisfies: $g(b_n)$ and $g(\beta_n)$ are rational complex for $n \in \mathbb{N}$. To that end, we introduce for each $n \in \mathbb{N}$ the set K_n as the set K of the second paragraph back with $r = (n+1)^{-1}$.† Clearly K_n is a compact subset of B . We choose γ_0, δ_0 so that

$$|\gamma_0 u_0(z) + \delta_0 w_0(z)| \leq 2^0, \quad z \in K_0,$$

and $h + (\gamma_0 u_0 + \delta_0 w_0)$ takes rational complex values at b_0 and β_0 . We proceed inductively, and having chosen γ_k, δ_k for $k = 0, \dots, n$, we take $\gamma_{n+1}, \delta_{n+1}$ so that

$$|\gamma_{n+1} u_{n+1}(z) + \delta_{n+1} w_{n+1}(z)| \leq 2^{-(n+1)}, \quad z \in K_{n+1},$$

and

$$\left[h + \sum_0^n (\gamma_k u_k + \delta_k w_k) \right] + (\gamma_{n+1} u_{n+1} + \delta_{n+1} w_{n+1})$$

takes rational complex values at b_{n+1} and β_{n+1} . With γ_n and δ_n so chosen the desired convergence properties as well as the stated requirement for g are assured.

Action of θ on constants. We proceed under the assumption that $\theta(i_A) = i_B$. The case where $\theta(i_A) = (-i)_B$ admits parallel treatment. Let $b = \lim_a \psi$. Let f be a nonconstant member of $A(A)$. Let $(a_n): \mathbb{N} \rightarrow A$ be such that $\lim_{n \rightarrow \infty} a_n = a$ and $f(a_n)$ is rational complex, $n \in \mathbb{N}$. The existence of (a_n) is assured by the open mapping property. By (2.2) we conclude

$$\theta(f)(b) = f(a). \quad (2.3)$$

Taking f nonconstant, $f(a) = 1$, and taking $c \in \mathbb{C}$, we see, on considering cf in place of f , that

$$[\theta(c_A)(b)][\theta(f)(b)] = cf(a),$$

so that $\theta(c_A) = c_B$. The complex constants are “preserved.” On appeal to

† Here, too, the Riemann surface situation would be treated differently, but the point is not serious.

(2.2) we reach the basic conclusion

$$\theta(f)(\psi(a)) = f(a), \quad (2.4)$$

$a \in A, f \in A(A)$. This relation permits us to complete the proof rapidly as far as the present setting is concerned. It suffices to take $\theta(f)$ as univalent and analytic on B ; ψ is then meromorphic on A for $\psi = \theta(f)^{-1} \circ f$. We take $\varphi = \psi^{-1}$. The proof is completed by noting that if $\varphi_1: B \rightarrow A$ satisfies $f \circ \varphi_1 = f \circ \varphi$, $f \in A(A)$, then $\varphi_1 = \varphi$, for if a_1 and a_2 are distinct points of A , there exists $f \in A(A)$ such that $f(a_1) \neq f(a_2)$. However to have a proof available also in a Riemann surface setting where such univalent functions may be lacking, we indicate an alternative attack. We take $g \in A(B)$ having a simple zero at $b = \lim_a \psi$ and no other zeros and conclude from (2.3) and (2.4) that $\psi(a) = b$. Consequently ψ is continuous on A . To show that ψ is meromorphic, we continue with g and take positive r so that $g|_{\Delta(b; r)}$ is univalent and thereupon positive ρ so that $\psi[\Delta(a; \rho)] \subset \Delta(b; r)$. Hence $\psi|_{\Delta(a; \rho)}$ is the composition of the inverse of $g|_{\Delta(b; r)}$ and $f|_{\Delta(a; \rho)}$ and consequently is meromorphic. The remaining details are readily supplied.

EXERCISES

- 2.1. Treat the case: $\theta(i_a) = (-i)_b$. It is convenient to consider the composition of functions that are either meromorphic or the conjugates of meromorphic functions.
 2.2. Show that if φ is a univalent meromorphic map of a region $B \subset \hat{\mathbb{C}}$ onto a region $A \subset \hat{\mathbb{C}}$, then $f \circ \varphi$ with domain $A(A)$ defines an isomorphism of $A(A)$ onto $A(B)$.

3. INFINITE PRODUCTS

It is well to keep in mind in the study of infinite products that when we deal with nonzero terms we want convergence to correlate with convergence of some infinite series whose terms are logarithms of the factors involved. The same goes for uniform convergence with factors not taking the value 0 or ∞ . Thanks to this introductory observation it will be seen that large sections of the initial parts of the subject make completely straightforward exercises at this point.

Definitions and elementary facts. Let there be given $(a_n): \mathbb{N} \rightarrow \mathbb{C}$. By the infinite product associated with (a_n) , denoted $\prod_0^\infty a_n$, is meant the sequence $(\prod_0^n a_k)_{n \in \mathbb{N}}$.† We say that $\prod_0^\infty a_n$ is *p-convergent* provided that for some $m \in \mathbb{N}$

† The notion may be introduced for more general algebraic structures where a multiplication is present.

the sequence $\prod_0^\infty a_{m+n}$ possesses a limit in $\mathbb{C} - \{0\}$.† In this case, $\prod_0^\infty a_n$ is used to denote

$$\lim_{n \rightarrow \infty} \prod_0^n a_k,$$

which exists in $\mathbb{C}$ and is termed the *product of the a_n , n ranging from 0 to ∞* . We say that $\prod_0^\infty a_n$ is *p-divergent* if it is not *p-convergent*. We say that $\prod_0^\infty a_n$ is *p-Cauchy* provided that for each positive ε there exists $m \in \mathbb{N}$ such that

$$\left| 1 - \prod_{p+1}^q a_k \right| < \varepsilon, \quad q > p \geq m.$$

The following theorem is easy to establish: $\prod_0^\infty a_n$ is *p-convergent* if and only if $\prod_0^\infty a_n$ is *p-Cauchy*. Consequently, if $\prod_0^\infty a_n$ is *p-convergent*, then $\lim_{n \rightarrow \infty} a_n = 1$. First suppose that $\prod_0^\infty a_n$ is *p-convergent* and that $m \in \mathbb{N}$ is such that $\prod_0^\infty a_{m+n} = b \in \mathbb{C} - \{0\}$. By the continuity of x/y with domain $\mathbb{C} \times (\mathbb{C} - \{0\})$ at (b, b) , we see that given positive ε there exists $\mu \in \mathbb{N}$ such that

$$\left| 1 - \frac{\prod_0^{q-m} a_{m+k}}{\prod_0^{p-m} a_{m+k}} \right| < \varepsilon$$

for $q > p \geq \mu + m$, that is,

$$\left| 1 - \prod_{p+1}^q a_k \right| < \varepsilon$$

for admitted p and q . Hence $\prod_0^\infty a_n$ is *p-Cauchy*. The fact that $\lim_{n \rightarrow \infty} a_n = 1$ is a consequence. Suppose now that $\prod_0^\infty a_n$ is *p-Cauchy*. We fix m such that

$$\left| 1 - \prod_{p+1}^q a_k \right| < 2^{-1}, \quad q > p \geq m.$$

Hence

$$\begin{aligned} \left| \prod_{m+1}^{q+(m+1)} a_k - \prod_{m+1}^{p+(m+1)} a_k \right| &= \left| \prod_{m+1}^{p+m+1} a_k \right| \left| 1 - \prod_{p+m+2}^{q+m+1} a_k \right| \\ &< \left(\frac{3}{2} \right) \left| 1 - \prod_{p+m+2}^{q+m+1} a_k \right|, \quad q > p \geq 0. \end{aligned}$$

We conclude that $(\prod_{m+1}^{p+(m+1)} a_k)_{p \in \mathbb{N}}$ is a Cauchy sequence and from the choice

† “ p ” is for “product.”

of m that $\prod_0^\infty a_{(m+1)+n}$ possesses a limit in $\overline{\Delta(1; \frac{1}{2})}$. Consequently $\prod_0^\infty a_n$ is p -convergent.

We say that $\prod_0^\infty a_n$ is *absolutely p -convergent* provided that $\prod_0^\infty (1 + |a_n - 1|)$ is p -convergent and that $\prod_0^\infty a_n$ is *absolutely p -Cauchy* provided that $\prod_0^\infty (1 + |a_n - 1|)$ is p -Cauchy. These conditions are, of course, equivalent to the conditions: the sequence

$$\left(\prod_0^n (1 + |a_k - 1|) \right)$$

possesses a limit in $\mathbf{R}$ (resp. is Cauchy).

We are now prepared to consider the first set of exercises on infinite products.

EXERCISES

3.1. Given $a_k \in \mathbf{C}$, $k = 1, \dots, n$. Show that

$$\begin{aligned} \sum_1^n |a_k|, \left| \prod_1^n (1 + a_k) - 1 \right| &\leq \prod_1^n (1 + |a_k|) - 1 \\ &\leq \exp\left(\sum_1^n |a_k|\right) - 1. \end{aligned}$$

3.2. $\prod_0^\infty a_n$ is absolutely p -convergent if and only if $\sum_0^\infty |a_n - 1|$ is convergent. Each of these conditions is equivalent to: for some $m \in \mathbf{N}$, $a_{m+n} \neq 0$ for all $n \in \mathbf{N}$ and $\sum_0^\infty |\text{Log } a_{m+n}|$ is convergent. If $\prod_0^\infty a_n$ is absolutely p -convergent, it is p -convergent.

3.3. $\prod_0^\infty a_n$ is p -convergent if and only if there exists $m \in \mathbf{N}$ such that $a_n \neq 0$, $n \geq m$, and $\sum_0^\infty \text{Log } a_{m+n}$ is convergent.

3.4. If $a_{2n} = (n+1)^{-1/2}$, $a_{2n+1} = -(n+1)^{-1/2}$, $n \in \mathbf{N}$, then $\prod_0^\infty (1 + a_n)$ p -diverges but $\sum_0^\infty a_n$ converges (however not absolutely). If

$$a_{2n} = -(n+1)^{-1/2}, \quad a_{2n+1} = (n+1)^{-1/2} + (n+1)^{-1}, \quad n \in \mathbf{N},$$

then $\sum_0^\infty a_n$ diverges but $\prod_0^\infty (1 + a_n)$ is p -convergent (however it is not absolutely p -convergent).

3.5. Evaluate

$$(a) \prod_0^\infty \left[1 + \frac{1}{(n+1)(n+3)} \right],$$

$$(b) \prod_0^\infty \left[1 - \frac{3}{(n+2)(n+4)} \right],$$

$$(c) \prod_0^\infty \frac{(n+2)^3 - 1}{(n+2)^3 + 1},$$

$$(d) \lim_{n \rightarrow \infty} \prod_0^n \frac{k^2 - k + A}{k^2 + k + A} \quad \text{where } k^2 + k + A \neq 0, \quad k \in \mathbb{N}.$$

3.6. Using $\sin \pi z = \pi z \prod_0^\infty [1 - (n+1)^{-2} z^2]$, represent $1/\pi$, $1/(\pi\sqrt{2})$, and $\sqrt{3}/(2\pi)$ as infinite products (cf. Ex. 1.8, this chapter). The case $z = \frac{1}{2}$ yields the Wallis formula for $\pi/2$.

3.7. Determine the sets of p -convergence (absolute p -convergence) of $\prod_0^\infty (1 - z^{n+1})$, $\prod_0^\infty (1 + z^{2^n})$, $\prod_0^\infty (1 + c_n z)$ given $\sum_0^\infty |c_n|$ convergent. Show that the products define functions that are analytic on the respective convergence sets.

3.8. Given $(a_n): \mathbb{N} \rightarrow \mathbb{C}$. Let ρ denote the supremum of

$$\left\{ |z| : \prod_0^\infty (1 + a_n z^n) \text{ } p\text{-converges} \right\}.$$

Show that ρ is the radius of convergence of $\sum_0^\infty a_n z^n$. Show that if $\rho > 0$, then the function given by $\prod_0^\infty (1 + a_n z^n)$ is analytic in $\Delta(0; \rho)$.

3.9. Show that when $|z| < 1$ we have

$$(a) \quad (1 - z)^{-1} = \prod_0^\infty (1 + z^{2^n}),$$

and

$$(b) \quad \left[\prod_0^\infty (1 - z^{2^{n+1}}) \right]^{-1} = \prod_0^\infty (1 + z^{n+1}).$$

3.10. Let $(p_n)_{n \in \mathbb{N}}$ denote the sequence of prime positive integers in increasing order: $p_0 = 2, p_1 = 3, \dots$. Then the domain of absolute p -convergence of

$$\prod_0^\infty [1 - \exp(-z \log p_n)]$$

is $\{\operatorname{Re} z > 1\}$ and for z in this set we have

$$\prod_0^\infty [1 - \exp(-z \log p_n)] = [\zeta(z)]^{-1}$$

where $\zeta(z) = \sum_1^\infty \exp(-z \log n)$. Hint: Use is to be made of the geometric series and of the multiplicative representation of a positive integer $\prod_0^\infty p_n^{v(n)}$, $v: \mathbb{N} \rightarrow \mathbb{N}$, $\{n: v(n) \neq 0\}$ finite.

3.11. Let $|a| < 1$. Let $f(z) = \prod_0^\infty (1 - a^{n+1} z)$. The function f is analytic on $\mathbb{C}$. Determine with the aid of the identity $f(z) = (1 - az)f(az)$ the coefficients of the power series expansion of f .

3.12. Let $0 < |a_n| < 1$, $n \in \mathbb{N}$, and suppose that $\sum_0^\infty (1 - |a_n|)$ is convergent. Show that

$$\prod_0^\infty \frac{\bar{a}_n}{|a_n|} \frac{a_n - z}{1 - \bar{a}_n z}$$

p -converges for $|z| < 1$ and show that the product represents on $\Delta(0; 1)$ a function f analytic on $\Delta(0; 1)$ and having modulus less than 1 and such that $\partial_f(z)$ is the number of elements of the set $\{n: a_n = z\}$. What happens when $|z| > 1$? Hint: If $|a| < 1$, $|z| = 1$, we have

$$\left| \frac{z-a}{1-\bar{a}z} \right| = |z| \left| \frac{z-a}{1-\bar{a}z} \right| = \left| \frac{1-\bar{a}z}{1-\bar{a}z} \right| = 1.$$

3.13. Let $(a_n): \mathbb{N} \rightarrow \mathbb{C} - \{0\}$, let $p \in \mathbb{N}$, and suppose that $\sum_0^\infty |a_n|^{-(p+1)}$ is convergent. Show that $\prod_0^\infty E_p(z/a_n)$ represents a function f analytic on $\mathbb{C}$ such that $\partial_f(z)$ is the number of elements in the set $\{n: a_n = z\}$. Show that the product is absolutely p -convergent using the hint of Ex. 1.3, this chapter. Show that $|f(z)| \leq \exp(A|z|^{p+1})$, A a positive number, using the result of the exercise just cited.

3.14. The gamma function. We consider the product of Ex. 3.13 with $a_n = -(n+1)$, $p = 1$ (for $p = 0$ the condition is not fulfilled). Let f denote the function represented by the product. Show that

$$(z+1)f(z+1) = f(z)e^{-z}$$

where

$$\gamma = \lim_{n \rightarrow \infty} \left[\left(\sum_{k=1}^n \frac{1}{k} \right) - \log(n+1) \right].$$

[γ is the *Euler constant*. It satisfies $0 < \gamma < 1$. The existence of the limit and the inequality just stated may be established with the aid of the representation

$$\log x = \int_1^x \frac{1}{t} dt, \quad 0 < x < +\infty.]$$

Hence introducing $g(z) = z f(z) e^{\gamma z}$, we see that $z g(z+1) = g(z)$. By the *gamma function* (of Euler), denoted Γ , is meant $1/g$. It satisfies

$$\Gamma(z+1) = z \Gamma(z)$$

(for what z ?).

Show that

$$(a) \quad g(z)g(-z) = -(z \sin \pi z)/\pi,$$

$$(b) \quad g(z)g(1-z) = (\sin \pi z)/\pi.$$

Calculate $\Gamma(\frac{1}{2})$. Show that

$$|g(it)|^2 = \frac{t \sinh(\pi t)}{\pi}, \quad t \in \mathbb{R}.$$

[Here $\sinh z = (e^z - e^{-z})/2$.] Show that

$$\frac{\prod_0^{n+1} (z+k)}{(n+1)! e^{z \log(n+1)}}$$

tends uniformly to g in $\mathbb{C}$. Determine with the aid of $z g(z+1) = g(z)$ the quantities $\text{res}(-n; \Gamma)$, $n \in \mathbb{N}$. Determine the 0-Laurent sequence of Γ'/Γ , of Γ , the terms of the

latter sequence being represented recursively with the aid of systems of linear equations depending on the index.

Uniformity questions. Let X be a set. We consider a sequence of functions $(f_n)_{n \in \mathbb{N}}$, each mapping X into $\mathbb{C}$. We say that $\prod_0^\infty f_n$ (which is the sequence $(\prod_0^n f_k)_{n \in \mathbb{N}}$) is *uniformly p -convergent on X* provided that $\prod_0^\infty f_n(x)$ is p -convergent for each $x \in X$ and that for each positive ε there exists $\mu \in \mathbb{N}$ such that

$$\left| 1 - \prod_0^\infty f_{m+n}(x) \right| < \varepsilon, \quad x \in X, \quad m \geq \mu.$$

We say that $\prod_0^\infty f_n$ is *uniformly p -Cauchy on X* provided that for each positive ε there exists $\mu \in \mathbb{N}$ such that

$$\left| 1 - \prod_{p+1}^q f_n(x) \right| < \varepsilon, \quad x \in X, \quad q > p \geq \mu.$$

When X is a Hausdorff space, we shall use the earlier "in" convention in the present context.

It is easy to check that (1) $\prod_0^\infty f_n$ is uniformly p -convergent on X if and only if $\prod_0^\infty f_n$ is uniformly p -Cauchy on X .

It is also easy to see that (2) $\prod_0^\infty f_n$ is uniformly p -convergent on X if and only if, for some $m \in \mathbb{N}$, the functions f_n do not take the value 0, $n \geq m$, and $\sum_0^\infty \text{Log} \circ f_{m+n}$ is uniformly convergent on X . The sufficiency is concluded by using the continuity of the exponential function at 0. The necessity may be concluded as follows. For some $\mu \in \mathbb{N}$, we have

$$\left| 1 - \prod_{p+1}^q f_n(x) \right| < 1, \quad x \in X, \quad q > p \geq \mu,$$

whence $f_n(x) \neq 0$, $x \in X$, $n \geq \mu + 1$. In addition, we have

$$\text{Log} \circ \left(\prod_{p+1}^q f_n \right) = \sum_{p+1}^q \text{Log} \circ f_n.$$

This may be seen by induction, since the assertion is trivially true for $q = p + 1$ and

$$\text{Log} \circ \left(\prod_{p+1}^q f_n \right) + \text{Log} \circ f_{q+1}$$

is a logarithm of $\prod_{p+1}^{q+1} f_n$ whose imaginary part takes values in $] -\pi, \pi[$ since the imaginary part of each of the two terms of the sum takes values in

$] -\pi/2, \pi/2[$. We thereupon conclude that, given positive $\varepsilon, \delta \in]0, 1[$ being such that $|\operatorname{Log} z| < \varepsilon$ for $|z - 1| < \delta$, if $\mu \in \mathbb{N}$ is now chosen so that

$$\left| 1 - \prod_{p=1}^q f_n(x) \right| < \delta, \quad x \in X, \quad q > p \geq \mu,$$

then

$$\left| \sum_0^\infty \operatorname{Log} \circ f_{p+n}(x) \right| \leq \varepsilon, \quad x \in X, \quad p \geq \mu + 1.$$

The necessity follows.

It is to be noted that if $\prod_0^\infty f_n$ is uniformly p -convergent on X , then for m sufficiently large the sequence $(\prod_0^n f_{m+k})_{n \in \mathbb{N}}$ tends uniformly on X (in the ordinary sense) to a function not taking the value 0. The example where $X =]0, 1]$, $f_n(x) = x^{(n+1)^{-2}}$, $n \in \mathbb{N}$, shows that the converse is false.

EXERCISES

3.15. Establish the last assertion.

3.16. If $\prod_0^\infty f_n$ is uniformly p -convergent on X , there exists $\mu \in \mathbb{N}$ such that $(\prod_0^n f_{m+k})_{n \in \mathbb{N}}$ converges uniformly on X (in the ordinary sense) to a function F_m bounded from 0 and ∞ , $m \geq \mu$. [That is, there exist α, β , $0 < \alpha < \beta < +\infty$, such that $\alpha \leq |F_m(x)| \leq \beta$, $x \in X$.] In the opposite direction, if there exists $m \in \mathbb{N}$ such that $(\prod_0^n f_{m+k})_{n \in \mathbb{N}}$ tends uniformly in the ordinary sense to a function bounded from 0, then $\prod_0^\infty f_n$ is uniformly p -convergent on X .

3.17. If for some $m \in \mathbb{N}$ the series $\sum_0^\infty |f_{m+n} - 1|$ converges uniformly on X , then $\prod_0^\infty f_n$ is uniformly p -convergent on X .

3.18. Test the products of Ex. 3.12 and 3.13, this section, for uniform and absolute p -convergence.

4. SUMS AND PRODUCTS

It frequently occurs in representations of functions by sums or products that the given enumeration of the terms is irrelevant. The best way to approach the matter is via a general notion of a sum or product for a function. It is not our object to develop the subject at length here, but rather to have ready a few convenient modes of expression and formulation.

Sums. Given a set A and $f: A \rightarrow \mathbb{C}$, for X finite $\subset A$, we define $s(X; f)$ as 0 when $X = \emptyset$ and as the common value of $\sum_0^n f(x_k)$ where $(x_k)_0^n$ is a univalent enumeration of X in case $X \neq \emptyset$. There is at most one $a \in \mathbb{C}$ such that for each

positive ε there exists B finite $\subset A$ such that

$$|s(X; f) - a| < \varepsilon,$$

X finite, $A \supset X \supset B$. When such a exists, we say that f is *summable* and term a the *sum* of f , denoting it $S(f)$. When A is finite, the notion of a sum agrees with that given earlier [Ex. 3.1, Ch. V]. Suppose that $f(A) \subset \mathbf{R}^+$. If in this case the set of $s(X; f)$ is bounded, f is summable and $S(f)$ is just the supremum of the set of $s(X; f)$. If in the same case f is summable, then the set of $s(X; f)$ is bounded.

Returning to the unrestricted case of complex-valued functions, we note that if $f, g: A \rightarrow \mathbf{C}$ are summable and $\alpha \in \mathbf{C}$, then $f + g$ and αf are summable and

$$S(f + g) = S(f) + S(g), \quad S(\alpha f) = \alpha S(f).$$

If $0 \leq f(x) \leq g(x)$, $x \in A$, and g is summable, then f is summable and $S(f) \leq S(g)$. Further, if $|f|$ is summable, then so is f and $|S(f)| \leq S(|f|)$. This may be seen as follows. Let $g = \operatorname{Re} f$, $h = \operatorname{Im} f$. Then $g^+ \leq |f|$, $-g^- \leq |f|$, $h^+ \leq |f|$, $-h^- \leq |f|$.† Since $g = g^+ + g^-$, g is summable. The same observation holds for h ; hence $f = g + ih$ is summable. The inequality follows on noting that $|s(X; f)| \leq S(|f|)$.

If f is real-valued and summable, then f^+ and f^- are summable and hence $|f|$ is. It suffices to take $\varepsilon = 1$ and to consider $s(X; f^+)$, $X \supset B$. If f is summable, then $\operatorname{Re} f$ and $\operatorname{Im} f$ are summable. Hence if f is summable, $|f|$ is summable. We conclude that if f is summable and B is now an arbitrary subset of A , then $f|B$ is summable. We denote $S(f|B)$ by $\sum_{x \in B} f(x)$. Here on occasion we shall write “ $P(x)$ ” for “ $x \in B$ ” where $P(x)$ is a sentential function such that $B = \{P(x)\}$. The same remark holds for products.

We recall that a set is said to be countable provided that it is finite or may be mapped univalently onto $\mathbf{N}$ and note that if f is summable, then $\{f(x) \neq 0\}$ is countable. If f is summable and $\{f(x) \neq 0\}$ is infinite, for every univalent (x_n) mapping $\mathbf{N}$ onto $\{f(x) \neq 0\}$, given positive ε there exists $m \in \mathbf{N}$ such that

$$\left| S(f) - \sum_0^n f(x_k) \right| < \varepsilon$$

for $n \geq m$.

What has been said for numerical-valued functions may be paraphrased for a map f from A into the family of complex-valued functions with a given domain B . The image of $a \in A$ will be denoted indicially f_a and the proximity is to be *uniform* in the sense of the Euclidean metric on $\mathbf{C}$. When a sum exists, we say that f is *uniformly summable on B* . The sum is denoted

† Here g^+ is the composition of x^+ , $x \in \hat{\mathbf{R}}$, with g ; g^- is similarly defined.

$\sum_{a \in A} f_a$. As in earlier considerations of convergence, in case B is a Hausdorff space, we also consider f *uniformly summable in B* , the definition being phrased in terms of the functions $f^K, f_a^K = f_a|K, K$ being compact $\subset B$. If f is uniformly summable on a Hausdorff space B and each f_a is continuous at $b \in B$, then $\sum_{a \in A} f_a$ is continuous at b . If B is an open subset of $\hat{\mathbb{C}}$, each f_a is analytic on B , and f is uniformly summable in B , then $\sum_{a \in A} f_a$ is analytic on B .

We may consider sums for meromorphic functions. Suppose that B is a region $\subset \hat{\mathbb{C}}$ and that f_a is meromorphic on B for each $a \in A$. Uniform summability of f in B is referred to the chordal metric. That is, f is termed χ -uniformly summable in B provided that there exists $g: B \rightarrow \hat{\mathbb{C}}$ such that for each positive ε and compact $K \subset B$ there exists X finite $\subset A$ such that for Z finite, $X \subset Z \subset A$, we have

$$\chi[s(Z; f)(z), g(z)] < \varepsilon, \quad z \in K.$$

Clearly such g is unique—it is termed *the χ -uniform sum of f in B* . Further, it follows from arguments drawn from our earlier studies concerning sequences of meromorphic functions that either g is the constant ∞_B or else is meromorphic on B . If for each $b \in B$ there exists positive r such that $\Delta(b; r) \subset B$, f_a is analytic in $\Delta(b; r)$ save for a in a finite subset X of A and g is uniformly summable in $\Delta(b; r)$, where $g_a = f_a| \Delta(b; r)$, $a \in A - X$, then $\sum_{a \in A} f_a$ is meromorphic on B .

EXERCISE

4.1. Validate the unproved assertions made thus far in this section.

Products. Parallel definitions are to be made for products. Starting with $f: A \rightarrow \mathbb{C}$ we introduce, as above, $p(\emptyset; f) = 1$ and $p(X; f) = \prod_0^n f(x_k)$. We say that f is *productible* provided that there exist $a \in \mathbb{C} - \{0\}$ and X finite $\subset A$ such that for each positive ε there exists Y finite $\subset A - X$ with the property that

$$|p(Z; f) - a| < \varepsilon$$

for Z finite, $Y \subset Z \subset A - X$. It is easy to see that $1 + |f - 1|$ is productible if and only if $|f - 1|$ is summable and that f is productible if $1 + |f - 1|$ is.† If f is productible, there exists unique $b \in \mathbb{C}$ such that for each positive ε there exists X finite $\subset A$ such that

$$|p(Y; f) - b| < \varepsilon$$

† Here “1” for “ 1_A .”

for Y finite, $X \subset Y \subset A$. We term b the *product* of f and denote it by $\prod_{a \in A} f(a)$.

EXERCISE

4.2. Validate the above assertions concerning products. To pass from “absolute productibility” ($1 + |f - 1|$ productible) to productibility, establish a “Cauchy” condition for $p(Z; f)$ and introduce $(Z_n)_{n \in \mathbb{N}}$ satisfying Z_n finite $\subset A$, $Z_n \subset Z_{n+1}$, $|p(Z - Z_n; f) - 1| < 2^{-(n+1)}$, Z finite, $Z_n \subset Z$. Does productibility imply absolute productibility?

Products with function factors. Again the definitions paraphrase those for sums. We allow (looking to our function-theoretic situation) f_a to take values in $\hat{\mathbb{C}}$. We say that f is *uniformly productible on B* provided that (1) there exists X finite $\subset A$ such that $f_a(b)$, $a \in A - X$, is productible for each $b \in B$, (2) for each positive ε there exists X finite $\subset A$ such that

$$\left| \prod_{a \in A - Y} f_a(b) - 1 \right| < \varepsilon,$$

$b \in B$, Y finite, $X \subset Y \subset A$. Again, “uniform in” is formulated via f_a^K .

In the meromorphic case uniform productibility in B (open $\subset \hat{\mathbb{C}}$) implies that there exists a unique function g meromorphic on B such that given positive ε and compact $K \subset B$ there exists Y finite $\subset A$ such that

$$\chi[p(Z; f)(z), g(z)] < \varepsilon$$

for $z \in K$, Z finite, $Y \subset Z \subset A$. We denote g by $\prod_{a \in A} f_a$. If no f_a is the constant zero, then $\prod_{a \in A} f_a$ is not.

EXERCISES

4.3. Develop the above remarks.

4.4. In the meromorphic situation, if $g = \prod_{a \in A} f_a$ is not the constant zero, then $\partial_g = \sum_{a \in A} \partial_{f_a}$.

4.5. In the meromorphic situation, if for each $b \in B$ there exists positive r such that $\Delta(b; r) \subset B$, f_a is analytic in $\Delta(b; r)$ save for a in a finite subset $X \subset A$ and g_a , $a \in A - X$, is uniformly summable on $\Delta(b; r)$ where $g_a = (|f_a - 1|) \Delta(b; r)$, then f is uniformly productible in B .

5. THE JENSEN FORMULA

Let f be meromorphic on a region A in $\hat{\mathbb{C}}$ but not the constant zero. Suppose that r is positive and such that $\Delta(0; r) \subset A$. The Jensen formula, which we

shall develop below, owes its importance to the fact that it relates in a way that turns out to be useful the following: $\partial_f | \Delta(0; r)$, the term of index $\partial_f(0)$ of the 0-Laurent sequence of f , $(\log |f|) | C(0; r)$. As preliminaries for developing the Jensen formula we shall want: (1) the Gauss mean-value formula for the real part of an analytic function, which follows from the Cauchy formula for the disk, on taking real parts, as well as from the theory of harmonic functions, (2) the formula (which we shall treat presently)

$$(2\pi)^{-1} \int_0^{2\pi} \log |re^{i\theta} - a| d\theta = \max\{\log r, \log |a|\}, \quad (5.1)$$

(3) the Möbius transformation L_a given by

$$L_a(z) = \frac{a - z}{1 - \bar{a}z}, \quad (5.2)$$

where $|a| < 1$. We recall that $|L_a(z)| = 1$, $|z| = 1$ (cf. Ex. 3.12, this chapter).

The formula (5.1). This can be verified very simply with the aid of the Gauss mean-value formula when $|a| \neq r$. Thus when $|a| > r$, we see that the left side of (5.1) is $\log |a|$ and we conclude (5.1) in this case. When $|a| < r$, the left side of (5.1) is

$$\log r + \frac{1}{2\pi} \int_0^{2\pi} \log \left| 1 - \frac{\bar{a}}{r} e^{i\theta} \right| d\theta,$$

and applying the Gauss mean-value formula, we conclude (5.1). When $|a| = r$, we see that the left side of (5.1) reduces to

$$\log r + \frac{1}{2\pi} \int_0^{2\pi} \log |1 - e^{i\theta}| d\theta.$$

The second term is seen to be 0 by applying the Cauchy integral theorem to the sum of the integrals of $[\text{Log}(1 - z)]/z$ along the paths

$$\overrightarrow{(e^{-i\alpha})(e^{i\alpha})}, \quad \gamma_\alpha(\theta) = e^{i\theta}, \quad \alpha \leq \theta \leq 2\pi - \alpha, \quad 0 < \alpha < \pi/2,$$

obtaining, on considering the limiting behavior,

$$\int_0^{2\pi} \log |1 - e^{i\theta}| d\theta = 0$$

(cf. Ex. 2.6, Ch. VII).

Let us turn to the development of the Jensen formula. We consider positive s such that $\overline{\Delta(0; s)} \subset A$ and $f(z) \neq 0, \infty$ when $r < |z| \leq s$. Let g be the non-vanishing analytic function on $\Delta(0; s)$ satisfying

$$f(z) = g(z) \prod_{|a| \leq r} \left[L_{a/s} \left(\frac{z}{s} \right) \right]^{\partial_f(a)} \quad (5.3)$$

Applying the Gauss mean-value theorem to $\log |g|$ we obtain

$$\log |g(0)| = \frac{1}{2\pi} \int_0^{2\pi} \log |g(re^{i\theta})| d\theta.$$

Now using

$$L_{a/s} \left(\frac{re^{i\theta}}{s} \right) = \frac{s(a - re^{i\theta})}{s^2 - \bar{a}re^{i\theta}}$$

and (5.1), we see that

$$\frac{1}{2\pi} \int_0^{2\pi} \log \left| L_{a/s} \left(\frac{re^{i\theta}}{s} \right) \right| d\theta = -\log s + \log r,$$

since $|a| \leq r$. We note that $\log |g(0)|$ is equal to

$$\frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{i\theta})| d\theta - \sum_{|a| \leq r} \frac{1}{2\pi} \int_0^{2\pi} \partial_f(a) \log \left| L_{a/s} \left(\frac{re^{i\theta}}{s} \right) \right| d\theta. \quad (5.4)$$

Letting c denote the $\partial_f(0)$ th term of the 0-Laurent sequence of f , from (5.3) we conclude

$$\log |c| = \log |g(0)| - \partial_f(0) \log s + \sum_{0 < |a| \leq r} \partial_f(a) \log \frac{|a|}{s}$$

and using (5.4), because of the arbitrariness of s , that

$$\log |c| = -\partial_f(0) \log r + \sum_{0 < |a| < r} \partial_f(a) \log \frac{|a|}{r} + \frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{i\theta})| d\theta, \quad (5.5)$$

a version of Jensen's formula.

We note that

$$\log \frac{r}{|a|} = \int_0^r \frac{\varphi_a(t)}{t} dt$$

where $\varphi_a(t) = 0$, $0 < t \leq |a|$, and $\varphi_a(t) = 1$, $|a| < t \leq r$, $0 < |a| < r$. Hence

$$\sum_{0 < |a| < r} \partial_f(a) \log \frac{r}{|a|} = \int_0^r \frac{\sum \partial_f^+(a) \varphi_a(t)}{t} dt + \int_0^r \frac{\sum \partial_f^-(a) \varphi_a(t)}{t} dt,$$

the sums on the right being taken over a satisfying $0 < |a| < r$. Hence

$$\sum_{0 < |a| < r} \partial_f(a) \log \frac{r}{|a|} = N(0; r) - N(\infty; r) - \partial_f(0) \log r, \quad (5.6)$$

where

$$N(0; r) = \int_0^r \frac{\nu(0; t) - \partial_f^+(0)}{t} dt + \partial_f^+(0) \log r,$$

and

$$N(\infty; r) = \int_0^r \frac{\nu(\infty; t) + \partial_f^-(0)}{t} dt - \partial_f^-(0) \log r,$$

$\nu(w; t)$ being the valence of $f | \Delta(0; t)$ at w . Using (5.5) and (5.6) we see that the Jensen formula may be written in the form

$$N(0; r) + \log |c| = N(\infty; r) + \frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{i\theta})| d\theta. \dagger \quad (5.7)$$

6. ENTIRE FUNCTIONS OF FINITE ORDER

We turn to the study of functions analytic on $\mathbb{C}$ —such functions are termed *entire* or *integral*. The case where they do not reduce to polynomials was discerned earlier; the term “transcendental entire” was then applied to them. Of particular interest in the present section will be entire functions that do not “grow too rapidly”—the concept will be treated precisely without delay—and the behavior of their divisors. We measure the growth of a function f analytic on $\Delta(0; \rho)$ by the behavior of $M(r) = \max_{C(0; r)} |f|$, $0 < r < \rho$. For an entire function f , we define the *order (of growth)* of f , denoted $\alpha(f)$, as

$$\limsup_{+\infty} \frac{\log^+ [\log^+ M(r)]}{\log r}. \ddagger \quad (6.1)$$

It is immediate that if $\alpha(f) < +\infty$, then for each positive $\beta > \alpha(f)$ we have

$$\log M(r) < r^\beta \quad (6.2)$$

for sufficiently large r . In the opposite direction, it is easy to see that the

† An elegant proof of the Jensen formula in the version (5.5) may be given with the aid of the theory of subharmonic functions, based on convexity properties of means of subharmonic functions. The version (5.7) brings us to the threshold of R. Nevanlinna's First Fundamental Theorem [51, 88, 89].

‡ If X is a topological space, $a \in X$ adheres to $X - \{a\}$, and F maps X or $X - \{a\}$ into $\hat{\mathbb{R}}$, by the *limit superior of F at a* , denoted $\limsup_a F$, is meant

$$\inf \{ \sup F(V - \{a\}) : V \text{ a neighborhood of } a \}.$$

In (6.1) we consider $r > 1$.

infimum of the set of positive β such that (6.2) is satisfied for large r is the order of f . To say that f *does not grow too rapidly* means simply that $\alpha(f) < +\infty$.

It is not hard to determine the order of some of the standard entire functions we have encountered. Thus for the exponential function $M(r) = e^r$ and the order is 1. For the sine from $|\sin z| \leq |\sin(i|z|)|$ we have $M(r) = (e^r - e^{-r})/2$ and the order is again 1. For a polynomial the order is 0. For $f(z) = \exp(e^z)$, $\alpha(f) = +\infty$. For

$$f(z) = \sum_0^{\infty} \frac{(-1)^k z^k}{(2k)!}, \quad \alpha(f) = \frac{1}{2}.$$

For

$$f(z) = \int_a^b e^{zt} d\mu(t),$$

where μ has finite variation, we have $|f(z)| \leq Ae^{B|z|}$, A and B being positive, and consequently at all events $\alpha(f) \leq 1$.†

EXERCISES

6.1. Show that for J_0 , the Bessel function of order 0, $M(r) = J_0(ir)$ and conclude from the power series expansion that

$$\frac{\sinh r}{r} < M(r) < e^r,$$

r positive, and that not only is $\alpha(J_0) = 1$ but also $\lim_{+\infty} r^{-1} \log M(r) = 1$.

6.2. Given $f(z) = \sum_0^{\infty} a_k z^k$ entire. Let $\mu(r) = \sum_0^{\infty} |a_k| r^k$. Show with the aid of the Cauchy inequalities that

$$M(r) \leq \mu(r) \leq M(\eta r) (1 - \eta^{-1})^{-1}, \quad 1 < \eta < +\infty,$$

and thereupon that

$$\alpha(f) = \limsup_{+\infty} \frac{\log^+ [\log^+ \mu(r)]}{\log r}.$$

The Jensen formula (5.7) yields useful information on the distribution of the zeros of an entire function f of finite order not the constant 0. Since $N(\infty; r) = 0$, we have, given $\beta > \alpha(f)$,

$$N(0; r) + \log |c| \leq r^\beta \quad (6.3)$$

† Far sharper results may be given with the aid of Phragmén–Lindelöf methods.

for large r . Since $v(0; t)$ is not decreasing and

$$N(0; 2r) \geq \int_r^{2r} \frac{v(0; t) - \partial_f^+(0)}{t} dt + \partial_f^+(0) \log(2r),$$

we have

$$(\log 2)[v(0; r) - \partial_f^+(0)] \leq N(0; 2r) - \partial_f^+(0) \log(2r),$$

and, combining this with (6.3), we conclude

$$\limsup_{+\infty} \frac{\log v(0; r)}{\log r} \leq \alpha(f). \quad (6.4)$$

Exponent of convergence. Let $\partial: \mathbf{C} \rightarrow \mathbf{Z}$ be such that $\{\partial(z) \neq 0\}$ clusters at no point of $\mathbf{C}$. By the *exponent of convergence* of ∂ , denoted $\kappa(\partial)$, is meant the infimum of the set of positive numbers β such that $|\partial(z)| |z|^{-\beta}$ with domain $\mathbf{C} - \{0\}$ is summable.† We shall show:

Theorem 6.1: *If f is entire, not the constant 0, then $\kappa(\partial_f) \leq \alpha(f)$.*

The case $\alpha(f) = +\infty$ is trivial. We put it aside. Let $\alpha(f) < \beta < \gamma < +\infty$. By (6.4) we have $v(0; r) \leq r^\beta$, r large. Noting that

$$\sum_{0 < |z| < r} \partial_f(z) |z|^{-\gamma} = \int_c^r t^{-\gamma} d[v(0; t) - \partial_f(0)], \quad r > c,$$

where c is a positive number such that $\partial_f(z) = 0$, $0 < |z| < c$, on integrating by parts, we see that

$$\sum_{0 < |z| < r} \partial_f(z) |z|^{-\gamma} = \frac{v(0; r) - \partial_f(0)}{r^\gamma} + \gamma \int_c^r \frac{v(0; t) - \partial_f(0)}{t^{\gamma+1}} dt, \quad r > c,$$

and, noting that $v(0; t) - \partial_f(0) \leq At^\beta$, $c \leq t < +\infty$, for some positive A , we conclude that

$$\sum_{0 < |z| < r} \partial_f(z) |z|^{-\gamma} \leq Ar^{\beta-\gamma} + \frac{\gamma}{\gamma-\beta} Ac^{\beta-\gamma}$$

and thereupon

$$\sum_{z \in \mathbf{C} - \{0\}} \partial_f(z) |z|^{-\gamma} \leq \frac{\gamma}{\gamma-\beta} Ac^{\beta-\gamma},$$

whence, given the arbitrariness of γ , we have the asserted inequality.

† We could, if we had wanted, have used $\partial(z) |z|^{-\beta}$ instead.

EXERCISE

6.3. Denoting $\sum_{0 < |z| < r} |\partial(z)|$ by $n(r)$, show that

$$\kappa(\partial) = \limsup_{r \rightarrow \infty} \frac{\log^+ n(r)}{\log r}.$$

The Weierstrass canonical product. Given an admitted ∂ with $\kappa(\partial) < +\infty$, we define the *genus* of ∂ as the minimum $\beta \in \mathbb{N}$ such that $|\partial(z)| |z|^{-(\beta+1)}$ with domain $\mathbb{C} - \{0\}$ is summable. Clearly, with p denoting the genus of ∂ we have

$$p \leq \kappa(\partial) \leq p + 1.$$

If $\kappa(\partial) \notin \mathbb{N}$, then p is the largest integer less than $\kappa(\partial)$. If $\kappa(\partial) \in \mathbb{N}$, then p is either $\kappa(\partial)$ or $\kappa(\partial) - 1$. The following examples show that both possibilities do arise. The first is ∂ taking the value 1 at each point of $\mathbb{N}$ and elsewhere the value 0. We have $\kappa(\partial) = 1$ and $p = 1$. The second is ∂ taking the value 1 at each point of

$$\{(n+1)[\log(n+1)]^{1+\varepsilon}; n \in \mathbb{N}\},$$

ε being a given positive number, and elsewhere the value 0. Here $\kappa(\partial) = 1$, $p = 0$.

For ∂ having a finite exponent of convergence we introduce

$$z^{\partial(0)} \prod_{a \in \mathbb{C} - \{0\}} [E_p(z/a)]^{\partial(a)}, \quad (6.5)$$

p being the genus of ∂ . Given r positive, for $|a| \geq 2r$ we have

$$|[E_p(z/a)]^{\partial(a)} - 1| \leq \exp(2|\partial(a)| |r/a|^{p+1}) - 1,$$

$|z| \leq r$, as we see using the equality $E_p(z) = \exp(-\sum_{k=1}^{\infty} z^k/k)$, $|z| < 1$. The uniform productibility of (6.5) follows.† We denote by P the meromorphic function defined by (6.5) and call it *the Weierstrass canonical product associated with ∂* . We see that $\partial_p = \partial$.

We continue with ∂ taking *nonnegative* values and see from Theorem 6.1 that $\kappa(\partial) \leq \alpha(P)$. We have however the more striking result:

Theorem 6.2. $\kappa(\partial) = \alpha(P)$.

PROOF: Let $\kappa(\partial) < \beta < +\infty$. Using the inequality of Ex. 1.3, this chapter, when $p \geq 1$, and proceeding directly when $p = 0$, we see that for some positive

† Recall: $e^x - 1 \leq xe^x$, $x \geq 0$.

As we have

$$\begin{aligned} |E_p(z)| &\leq e^{A|z|^{p+1}}, & |z| \leq 1, \\ &\leq e^{A|z|^\beta}, & |z| > 1. \end{aligned}$$

Hence

$$\begin{aligned} \log M(r) &\leq \partial(0) \log r + A \sum_{0 < |a| < r} \partial(a) \left| \frac{r}{a} \right|^\beta + A \sum_{|a| \geq r} \partial(a) \left| \frac{r}{a} \right|^{p+1} \\ &= \partial(0) \log r + Ar^\beta \int_0^r \frac{dn(t)}{t^\beta} + Ar^{p+1} \int_r^{+\infty} \frac{dn(t)}{t^{p+1}}. \end{aligned} \quad (6.6)$$

Here $n(r) = \sum_{0 < |z| < r} \partial(z)$ as in Ex. 6.3, this section.

To proceed we distinguish two cases. (1) $\kappa(\partial) = p + 1$. From (6.6) we see that there exists positive B such that $\log M(r) < Br^\beta$ for large r . We conclude, given the arbitrariness of β , that $\kappa(\partial) \geq \alpha(P)$. Hence the asserted equality holds. (2) $\kappa(\partial) < p + 1$. We take $\beta < p + 1$. By Ex. 6.3, this section, we have $n(r) < r^\beta$ for large r and hence, using integration by parts,

$$r^{p+1} \int_r^{+\infty} \frac{dn(t)}{t^{p+1}} < \frac{p+1}{(p+1) - \beta} r^\beta$$

for large r and again we conclude the asserted equality with the aid of (6.6).†

We are now in a position to obtain the important representation theorem of Hadamard for entire functions of finite order. It will be convenient to consider first an entire function f of finite order having no zeros. Let g denote an analytic logarithm of f and let $h = \operatorname{Re} g$. Let q denote the largest integer not exceeding $\alpha(f)$. Using formula (5.6) of Ch. IV applied to $g(rz)$, r positive, we conclude taking the $(q+1)$ st derivative that

$$g^{(q+1)}(z) = \frac{(q+1)!}{2\pi r^{q+1}} \int_0^{2\pi} h(re^{i\theta}) \frac{2e^{i\theta}}{[e^{i\theta} - (z/r)]^{q+2}} d\theta, \quad |z| < r. \quad (6.7)$$

Now using $|a| = 2a^+ - a$, $a \in \mathbf{R}$, and the Gauss mean-value theorem, we conclude that

$$\frac{1}{2\pi} \int_0^{2\pi} |h(re^{i\theta})| d\theta = \frac{1}{\pi} \int_0^{2\pi} h^+(re^{i\theta}) d\theta - h(0). \quad (6.8)$$

With β satisfying $\alpha(f) < \beta < q + 1$, for r large we have

$$h^+(re^{i\theta}) = \log^+ |f(re^{i\theta})| < r^\beta,$$

† More refined estimates may be obtained for $\log M(r)$ when $\partial(z) |z|^{-\kappa(\partial)}$, $z \in \mathbf{C} - \{0\}$, is summable.

and from (6.7) and (6.8)

$$|g^{(q+1)}(z)| \leq (q+1)! \frac{2}{[1 - (|z|/r)]^{q+2}} \cdot \frac{2r^\beta + |h(0)|}{r^{q+1}},$$

$|z| < r$. Hence $g^{(q+1)}$ is the constant 0 and g is a polynomial of degree $\leq q$. Since $\alpha(f)$ does not exceed the positive part of the degree of g , we conclude that $\alpha(f) = q$.†

To continue, suppose that f is an entire function of finite order, not the constant zero. Let P denote the Weierstrass canonical product associated with ∂_f and let $h = f/P$. We have the inequality

$$\alpha(h) \leq \alpha(f). \quad (6.9)$$

To see this, we note that

$$\log^+ |h| \leq \log^+ |f| + \log^+ |1/P|. \quad (6.10)$$

Applying the Jensen formula (5.7) to P we have

$$\int_0^{2\pi} \log^+ \left| \frac{1}{P(re^{i\theta})} \right| d\theta \leq \int_0^{2\pi} \log^+ |P(re^{i\theta})| d\theta \quad (6.11)$$

for $r \geq 1$. Applying the Poisson integral formula (5.7), Ch. IV, to $\log |h(2rz)|$, we have

$$\log |h(z)| = \frac{1}{2\pi} \int_0^{2\pi} \log |h(2re^{i\theta})| \operatorname{Re} \left(\frac{2re^{i\theta} + z}{2re^{i\theta} - z} \right) d\theta, \quad |z| < 2r.$$

Hence using the notation $M(r; F)$ to denote $\max_{C(0; r)} |F|$ —in order to indicate the function in question—and noting the inequality

$$\operatorname{Re} \left(\frac{2re^{i\theta} + z}{2re^{i\theta} - z} \right) \leq \frac{2r + |z|}{2r - |z|}, \quad |z| < 2r,$$

on appeal to (6.10) and (6.11), we obtain

$$\begin{aligned} \log M(r; h) &\leq \frac{3}{2\pi} \int_0^{2\pi} \log^+ |h(2re^{i\theta})| d\theta \\ &\leq \frac{3}{2\pi} \left[\int_0^{2\pi} \log^+ |f(2re^{i\theta})| d\theta + \int_0^{2\pi} \log^+ |P(2re^{i\theta})| d\theta \right] \\ &\leq 3[\log^+ M(2r; f) + \log^+ M(2r; P)], \quad r \geq \frac{1}{2}. \end{aligned}$$

We conclude on appeal to Theorems 6.1 and 6.2 that $\alpha(h) \leq \alpha(f)$. We are led, using what we have learned about entire functions of finite order having no zeros, to the theorem of Hadamard, which follows.

† The degree of g is taken as $\sup\{k: a_k \neq 0\}$ where $g(z) = \sum_0^\infty a_k z^k$.

Theorem 6.3: Given f entire of finite order, not the constant 0. Then $f = (\exp \circ Q)P$, where P is the Weierstrass canonical product associated with ∂_f and Q is a polynomial of degree $\leq \alpha(f)$. Q is determined up to an additive constant taking the value $2\pi in$, $n \in \mathbb{Z}$.

The genus of an entire function f of finite order is defined as the maximum of the genus of ∂_f and the order of $\exp \circ Q$.

The theorem of Hadamard has many important applications. We shall learn some in the exercise set that follows.

EXERCISES

6.4. Let f and g be entire functions. Show that $\alpha(f+g), \alpha(fg) \leq \max\{\alpha(f), \alpha(g)\}$ and that, if g is not the constant 0 and f/g is entire, $\alpha(f/g) \leq \max\{\alpha(f), \alpha(g)\}$.

Show that $M(r; f) \leq r M(r; f') + |f(0)|$ and that $\overrightarrow{M}(r; f') \leq 2 M(2r; f)/r$ and conclude that $\alpha(f') = \alpha(f)$. Hint: Integrate f' along $\vec{0z}$, $|z| = r$, to obtain the first inequality and apply the integral formula for f' for the second inequality.

6.5. Given $\beta \in \mathbb{R}^+$ and an admitted ∂ taking nonnegative values. Show that a necessary and sufficient condition that there exist f entire, $\alpha(f) = \beta$, $\partial_f = \partial$, is that $\kappa(\partial) = \beta$ when $\beta \in \mathbb{R}^+ - \mathbb{N}$ and $\kappa(\partial) \leq \beta$ when $\beta \in \mathbb{N}$. Characterize the set of entire f satisfying $\alpha(f) = \beta$.

6.6. If f is entire and of finite order and $\alpha(f) \notin \mathbb{N}$, then for each $w \in \mathbb{C}$ the function $g = f - w_{\mathbb{C}}$ satisfies $\alpha(g) = \kappa(\partial_g) = \alpha(f)$. If f is transcendental entire and $\alpha(f) = 0$, then g has infinitely many zeros.

6.7. If f is entire, $\alpha(f) \leq 1$, and f is even (that is, $f(-z) = f(z)$) but f is not the constant 0, then Q of Theorem 6.3 is constant. In the special case where $\alpha(f) = 1$, f has infinitely many zeros.

6.8. Let P denote the Weierstrass canonical product associated with the divisor of $\sin \pi z$. Using the evenness of $(\sin \pi z)/P(z)$, conclude that $\sin \pi z = \pi P(z)$.

6.9. Let $\partial: \mathbb{C} \rightarrow \mathbb{Z}$ satisfy $\kappa(\partial) < +\infty$ and let p be the genus of ∂ . Given $a \in \mathbb{C} - \{0\}$ let

$$M_a(z) = \frac{1}{z-a} + \sum_{k=1}^{p-1} \frac{z^k}{a^{k+1}},$$

and let $M_0(z) = 1/z$. Show that $\partial(a)M_a$ (with domain $\mathbb{C}$) is uniformly χ -summable in $\mathbb{C}$, the sum being meromorphic on $\mathbb{C}$ and having principal part at a given by $\partial(a)(z-a)^{-1}$. Formulate and prove a generalization where ∂ is replaced by a function taking values in $\mathbb{C}$, the preimage of $\mathbb{C} - \{0\}$ however clustering at no point of $\mathbb{C}$.

6.10. The present exercise affords a brief glimpse of some of the important functions arising in the theory of elliptic functions. Let $\alpha, \beta \in \mathbb{C} - \{0\}$ and satisfy $\text{Im}(\alpha^{-1}\beta) \neq 0$. Let Ω denote the additive subgroup of $\mathbb{C}$ generated by $\{\alpha, \beta\}$, that is, Ω consists of the complex numbers of the form $m\alpha + n\beta$, $m, n \in \mathbb{Z}$.

Show that ∂ taking the value 1 on Ω and the value 0 on $\mathbb{C} - \Omega$ has exponent of convergence and genus both equal to 2. Hint: Show that there exist A, B , $0 < A < B < +\infty$, such that

$$A(m^2 + n^2)^{1/2} \leq |m\alpha + n\beta| \leq B(m^2 + n^2)^{1/2}.$$

Estimate for each $m \in \mathbb{Z} - \{0\}$ the sum $\sum_0^\infty (m^2 + n^2)^{-a/2}$ with the aid of

$$\int_0^{+\infty} (m^2 + x^2)^{-a/2} dx$$

and

$$\int_0^{+\infty} (m^2 + x^2)^{-a/2} dx + |m|^{-a}$$

when $a > 1$.

Let σ denote the Weierstrass canonical product associated with ∂ . [It is termed *the σ -function associated with Ω* .] Show that σ is odd, that is, $\sigma(-z) = \sigma(z)$. Show that

$$\sigma(z + \alpha) = \sigma(z) \exp[Q_\alpha(z)]$$

where Q_α is a polynomial of degree at most 1 and that a corresponding statement holds with β replacing α . Hint: We know that the above statement holds for some polynomial Q_ω of degree at most 2 when $\omega \in \Omega$ replaces α . Show that

$$\exp[Q_\alpha(z - \alpha) + Q_{-\alpha}(z)] = 1,$$

and from the fact that σ is odd that

$$\exp[Q_\alpha(-z)] = \exp[Q_{-\alpha}(z)],$$

and conclude that the degree of Q_α is not greater than 1.

We introduce *the ζ -function associated with Ω* (not to be confused with the Riemann ζ -function) as $\zeta = \sigma'/\sigma$, and *the Weierstrass p -function associated with Ω* as $p = -\zeta'$. Let $\tau_\alpha(z) = z + \alpha$, $\tau_\beta(z) = z + \beta$. Show that $\zeta \circ \tau_\alpha - \zeta$ is the constant taking the value $2\zeta(\alpha/2)$ and $\zeta \circ \tau_\beta - \zeta$ is the constant taking the value $2\zeta(\beta/2)$. Show that $p \circ \tau_\alpha = p \circ \tau_\beta = p$. *Calculate with the aid of the residue theorem

$$\sum_1^4 \int_{\gamma_k} \zeta dz$$

where

$$\begin{aligned} \gamma_1 &= \overrightarrow{\left(\frac{-\alpha-\beta}{2}\right)\left(\frac{\alpha-\beta}{2}\right)}, & \gamma_2 &= \overrightarrow{\left(\frac{\alpha-\beta}{2}\right)\left(\frac{\alpha+\beta}{2}\right)}, \\ \gamma_3 &= \overrightarrow{\left(\frac{\alpha+\beta}{2}\right)\left(\frac{-\alpha+\beta}{2}\right)}, & \gamma_4 &= \overrightarrow{\left(\frac{-\alpha+\beta}{2}\right)\left(\frac{-\alpha-\beta}{2}\right)} \end{aligned}$$

and conclude the Legendre relation

$$\beta \zeta\left(\frac{\alpha}{2}\right) - \alpha \zeta\left(\frac{\beta}{2}\right) = \pi i \operatorname{sg}[\operatorname{Im}(\alpha^{-1}\beta)]$$

and hence that at least one of the polynomials Q_α , Q_β has degree 1. Show that when $\alpha = 1$ the function assigning to β the value $\zeta(1/2)$ is analytic on $\{\operatorname{Im} z \neq 0\}$ and that the same holds with $\zeta(1/2)$ replaced by $\zeta(\beta/2)$. Prove a similar result for the 0-Laurent coefficients of ζ and of σ .

6.11. A theorem of Laguerre. Let f be an entire function of finite order, not the constant 0. Suppose that the genus of f does not exceed 1, that $f(\mathbf{R}) \subset \mathbf{R}$, and that the zeros of f are all real. Show with the aid of the Hadamard representation theorem, using the logarithmic derivative of f , that the zeros of f' are all real. Note that the theorem fails for entire functions of finite order with genus exceeding 1. Consider, for example, $f(z) = ze^{z^2/2}$, for which $f'(z) = (1 + z^2)e^{z^2/2}$.

7. BOUNDED ANALYTIC FUNCTIONS

We now turn to the problem of characterizing the divisor of a function analytic on $\Delta(0; 1)$, bounded, and not the constant 0. The Jensen formula (5.7), this chapter, leads rapidly to a necessary condition. Let f be a function satisfying the stated conditions. We suppose, as we may without loss of generality, that $|f(z)| < 1$, $|z| < 1$. By (5.7) we have $N(0; r) \leq -\log |c|$. We conclude

$$\lim_1 [\nu(0; r) - \partial_f(0)] \log \frac{1}{r} = 0.$$

Applying an integration by parts to the formula defining $N(0; r)$ and using the result of the preceding sentence we obtain the inequality

$$\int_0^1 \log \frac{1}{t} d[\nu(0; t) - \partial_f(0)] \leq -\log |c|,$$

which may be recast

$$\sum_{0 < |z| < 1} \partial_f(z) \log \frac{1}{|z|} \leq -\log |c|.$$

Hence a *necessary condition* for $\partial: \Delta(0; 1) \rightarrow \mathbf{N}$ to be the divisor of a function analytic on $\Delta(0; 1)$, bounded, and not the constant 0 is that $-\partial(z) \log |z|$, with domain $\Delta(0; 1) - \{0\}$, be summable. Given ε , $0 < \varepsilon < 1$, the set of values taken by $(-\log r)/(1 - r)$ on $[\varepsilon, 1[$ is bounded above and its infimum is 1. We see that the following condition, the so-called *Blaschke condition*, is equivalent to the necessary condition on ∂ just stated: $\partial(z) (1 - |z|)$, with domain $\Delta(0; 1)$, is summable.

We shall approach the problem also in a second way, one by which we shall be able to proceed to the *sufficiency* aspect of the question thanks to the tools that will be introduced. We shall continue to use the notation L_a of §5, this chapter, but now with the understanding that L_a is the function with domain $\Delta(0; 1)$ given by (5.2). Suppose that E is a finite subset of $\Delta(0; 1)$. We see that

$$g = \frac{f}{\prod_{a \in E} (L_a)^{\partial_f(a)}} \quad (7.1)$$

is analytic on $\Delta(0; 1)$. It is routine to see that

$$|L_{|a|}(|z|)| \leq |L_a(z)| \leq L_{|a|}(-|z|).$$

Indeed it suffices to consider $|L_a(z)|^2$ and to use the very elementary observation that $(A + x)/(B + x)$ is increasing on $\{x \geq -A\}$, where $A < B$. Using the notation for the maximum modulus on $C(0; r)$ introduced in §6, this chapter, we see that

$$M(r; g) \leq \frac{M(r; f)}{\prod_{a \in E} |L_{|a|}(r)|^{\partial_f(a)}}, \quad 0 < r < 1,$$

and we conclude $\lim_1 M(r; g) \leq 1$. It is to be noted that by the boundary maximum principle $M(r; g)$ is never decreasing. We conclude that $|g(z)| \leq 1$, $|z| < 1$. If $|g|$ attains the value 1 at some point of $\Delta(0; 1)$, then g is a constant taking a value of modulus 1. We see that if X is a finite subset of $\Delta(0; 1) - \{0\}$, on letting E in (7.1) be $X \cup \{0\}$, we have

$$1 \geq \frac{|c|}{\prod_{a \in X} |a|^{\partial_f(a)}},$$

where c is the 0-Laurent coefficient of f of index $\partial_f(0)$. Consequently,

$$\prod_{0 < |z| < 1} |z|^{-\partial_f(z)} \leq |c|^{-1}.$$

This condition goes into the one given by the Jensen formula on taking logarithms. The two conditions are equivalent.

Proceeding in the opposite direction, we start with $\partial: \Delta(0; 1) \rightarrow \mathbb{N}$ satisfying the Blaschke condition. We introduce the product

$$\prod_{|a| < 1} [\varepsilon(a) L_a]^{\partial(a)}, \quad (7.2)$$

where $\varepsilon(0) = -1$ and $\varepsilon(a) = \bar{a}/|a|$, $0 < |a| < 1$. In essence, the product was studied in Ex. 3.12, this chapter, save for the question of the independence of the arrangement of terms. However, we shall give an autonomous treatment here. It suffices to note that

$$\begin{aligned} |\varepsilon(a) L_a(z) - 1| &= (1 - |a|) \left| \frac{|a| + \bar{a}z}{|a|(1 - \bar{a}z)} \right| \\ &\leq (1 - |a|) \left(\frac{1 + |z|}{1 - |z|} \right), \end{aligned}$$

when $0 < |a| < 1$, $|z| < 1$, in order to see that $[\varepsilon(a) L_a]^{\partial(a)}$ with domain $\Delta(0; 1)$ is uniformly productible in $\Delta(0; 1)$. The product (7.2) is an analytic

function on $\Delta(0; 1)$ taking values of modulus at most 1 whose divisor is precisely ∂ . It is termed the *Blaschke product associated with ∂* . We have the following theorem, which settles the problem of characterizing the divisor of a function analytic on $\Delta(0; 1)$ and bounded which is not the constant 0.

Theorem 7.1: *A necessary and sufficient condition that $\partial: \Delta(0; 1) \rightarrow \mathbf{N}$ be the divisor of a function analytic on $\Delta(0; 1)$ and bounded which is not the constant 0 is that ∂ satisfy the Blaschke condition.*

As a by-product of the proof we obtain: *If f is analytic on $\Delta(0; 1)$, not the constant 0, and if $|f(z)| \leq 1$, $|z| < 1$, then f/B is a function analytic on $\Delta(0; 1)$, omitting the value 0, whose modulus takes values not exceeding 1, B being the Blaschke product associated with ∂_f .* It suffices to note that if E is a finite subset of $\Delta(0; 1)$, then

$$\frac{f}{\prod_{a \in E} [\varepsilon(a) L_a]^{\partial_f(a)}}$$

is analytic on $\Delta(0; 1)$ and its modulus takes values no greater than 1. Hence f/B , which is analytic on $\Delta(0; 1)$ and omits the value 0, takes values of modulus at most 1 on $\{\partial_f(z) = 0\}$, and consequently, by continuity, on $\Delta(0; 1)$. More generally, when the condition $|f(z)| \leq 1$, $|z| < 1$, is replaced by the boundedness of f , the suprema of the set of values of $|f|$ and of $|f/B|$ are the same. It suffices to consider αf where α is the reciprocal of the supremum of the set of values of $|f|$.

EXERCISES

7.1. The lemma of Schwarz. Let f be analytic on $\Delta(0; 1)$, have modulus taking values less than 1, and satisfy $f(0) = 0$. Then $|f(z)| \leq |z|$, $|z| < 1$, and $|f'(0)| \leq 1$. If $|f(a)| = |a|$ for some $a \in \Delta(0; 1) - \{0\}$ or $|f'(0)| = 1$, then $f(z) = \eta z$ where $\eta \in C(0; 1)$. [The fact that if $f(z) = \eta z$, then $|f'(0)| = 1$ and $|f(z)| = |z|$, $|z| < 1$ is trivial.] Hint: Consider f/L_0 .

7.2. Conclude with the aid of the lemma of Schwarz that f is a univalent function analytic on $\Delta(0; 1)$ mapping $\Delta(0; 1)$ onto itself and satisfying $f(0) = 0$ if and only if $f(z)$ is of the form ηz , $\eta \in C(0; 1)$. Hint: For necessity start from $f^{-1}[f(z)] = z$ and deduce that $|z| = |f(z)|$ (also we may deduce $|f'(0)| = 1$).

Show that f satisfies the above condition save possibly the requirement $f(0) = 0$ if and only if $f(z)$ is of the form $L_a(\eta z)$, $\eta \in C(0; 1)$. Hint: Consider $L_a \circ f$, $a = f(0)$.

7.3. Lemma of Schwarz–Pick. Given $f: \Delta(0; 1) \rightarrow \Delta(0; 1)$, f analytic on $\Delta(0; 1)$. Let $|a| < 1$. Show that

$$\frac{|f'(a)|}{1 - |f(a)|^2} \leq \frac{1}{1 - |a|^2}.$$

equality holding if and only if f is a univalent map of $\Delta(0; 1)$ onto itself. Hint: Apply the chain rule to $L_b \circ f \circ L_a$, where $b = f(a)$.

7.4. For this exercise L_a is to be construed as the Möbius transformation satisfying $L_a(z) = (a - z)/(1 - \bar{a}z)$, $z \neq \infty$. Here $|a| < 1$. We suppose that $\partial: \Delta(0; 1) \rightarrow \mathbb{N}$ satisfies the Blaschke condition. Show that $[\varepsilon(a) L_a]^{\partial(a)}$ with domain $\Delta(0; 1)$ is uniformly productible in the complement with respect to $\hat{C}$ of the set of cluster points of $\{\partial(z) \neq 0\}$. Show that for $|z| > 1$ the divisor of the product at z is $-\partial(1/\bar{z})$. Show that the product is analytic at each point of $C(0; 1)$ not a cluster point of $\{\partial(z) \neq 0\}$.

Functions analytic on a half-plane and bounded. For certain purposes it is important to have conditions characterizing the distribution of the zeros of a function analytic on a half-plane and bounded. In theory, we may treat even more general situations when we can reduce the problem to that treated for the unit disk. Suppose that f is meromorphic on a region $A \subset \hat{C}$ and that $\varphi: B \rightarrow A$ where B is a region $\subset \hat{C}$ and φ is meromorphic on B but is not constant. Then by Ex. 4.14, Ch. VI, we have

$$\partial_{f \circ \varphi}(z) = [\partial_f \circ \varphi(z)] n(z; \varphi), \quad z \in B.$$

When φ is univalent, $n(z; \varphi) = 1$, $z \in B$. Let us now specialize B taking it to be $\Delta(0; 1)$ and suppose that φ maps $\Delta(0; 1)$ univalently onto A . From $\partial_{f \circ \varphi} = \partial_f \circ \varphi$, we see that ∂ is a divisor of a function analytic on A , bounded, and not the constant 0 if and only if $\partial \circ \varphi$ satisfies the Blaschke condition or, equivalently, if and only if $\partial(z) [1 - |\varphi^{-1}(z)|]$ with domain A is summable.

In the special case where $A = \{\operatorname{Re} z > 0\}$, we are in a position to exhibit a map φ explicitly. Indeed, it is easily checked that

$$\psi(z) = \frac{1 - z}{1 + z}, \quad \operatorname{Re} z > 0,$$

is a univalent map of $\{\operatorname{Re} z > 0\}$ onto $\Delta(0; 1)$. We take $\varphi = \psi^{-1}$. Now

$$1 - |\psi(z)| = \frac{|1 + z|^2 - |1 - z|^2}{|1 + z|(|1 + z| + |1 - z|)},$$

and, since $\operatorname{Re} z > 0$, we have

$$1 + |z|^2 < |1 + z|^2 + |1 - z|^2 < 3(1 + |z|^2).$$

Hence

$$\frac{4 \operatorname{Re} z}{3(1 + |z|^2)} < 1 - |\psi(z)| < \frac{4 \operatorname{Re} z}{1 + |z|^2}.$$

Consequently, ∂ is the divisor of a function analytic on $\{\operatorname{Re} z > 0\}$, bounded, not the constant 0, if and only if $\partial(z) (\operatorname{Re} z)(1 + |z|^2)^{-1}$ with domain $\{\operatorname{Re} z > 0\}$ is summable.

EXERCISES

7.5. Given $\operatorname{Re} a$ positive. Show that $\lambda_a(z) = (a - z)/(\bar{a} + z)$ with domain $\{\operatorname{Re} z > 0\}$ is a univalent map of $\{\operatorname{Re} z > 0\}$ onto $\Delta(0; 1)$. Show that f is a univalent function analytic on $\{\operatorname{Re} z > 0\}$ with image $\Delta(0; 1)$ satisfying $f(a) = 0$ if and only if $f = \eta\lambda_a$, $\eta \in C(0; 1)$. State and prove an analog of the lemma of Schwarz for g analytic on $\{\operatorname{Re} z > 0\}$ whose modulus takes values less than 1, $g(a)$ being 0.

7.6. Given $\operatorname{Re} b$ positive. Suppose that Φ is the set of f analytic on $\Delta(0; 1)$ satisfying $\operatorname{Re} f(z) > 0$, $|z| < 1$, and $f(0) = b$. Show that $\{|f'(0)| : f \in \Phi\}$ has a maximum; determine its value and the members of Φ which yield the maximum. Hint: Consider $\lambda_b \circ f$ and apply the lemma of Schwarz.

7.7. Suppose that ∂ mapping $\{\operatorname{Re} z > 0\}$ into $\mathbb{N}$ satisfies the condition that $\partial(z) = 0$ for z small and the condition of the text: $\partial(z)(\operatorname{Re} z)(1 + |z|^2)^{-1}$ summable. Show that $[(\bar{a}/a)\lambda_a]^{\partial(a)}$ with domain $\{\operatorname{Re} a > 0\}$ is uniformly productible in $\{\operatorname{Re} z > 0\}$. Prove a corresponding result when $\partial(z) = 0$ for z large, -1 replacing $\bar{a}/a$.

7.8. Show directly that if f is analytic on $\{\operatorname{Re} z > 0\}$, takes values of modulus less than 1, and is not the constant 0, then when E is a finite subset of $\{\operatorname{Re} z > 0\}$, $f/\prod_{a \in E} (\lambda_a)^{\partial f(a)}$ is analytic on $\{\operatorname{Re} z > 0\}$ and takes values of modulus at most 1. Conclude that the divisor condition of the text cited in the preceding exercise characterizes the divisors of the functions analytic on $\{\operatorname{Re} z > 0\}$ which are bounded and different from the constant 0.